12 kW Wind Turbine for Iceland: Generator Manufacturing Instructions (P1)

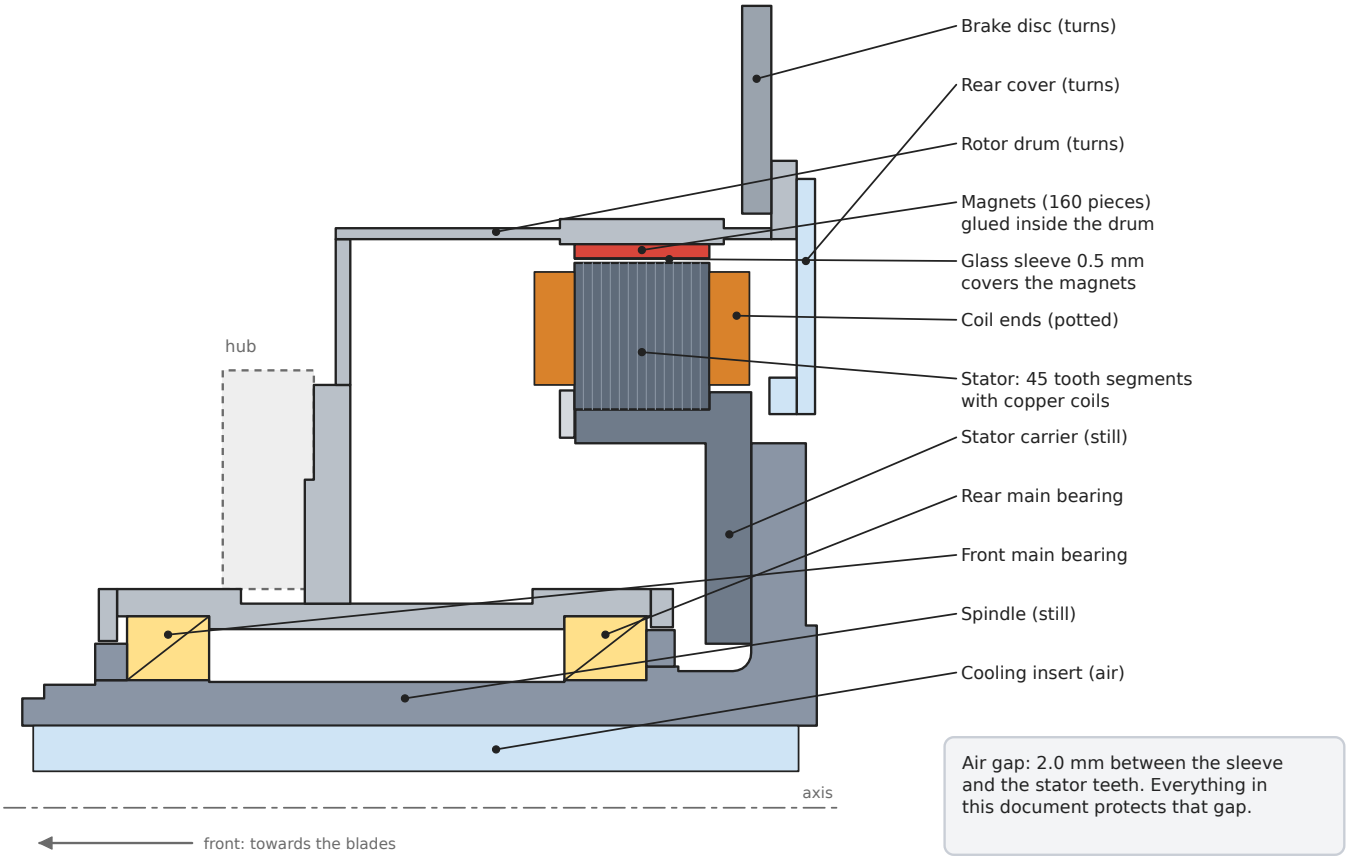
Status: **Draft A.1 (pictures added), not yet process-qualified** · 2026-11-04 Owner: Category 10 (Manufacturing and quality) with 4 (Generator) and 11 (Testing) Builds on: 06 – Generator round 2 (frozen electromagnetic design), 14 – Nacelle prototype P1 (parts, fits, air gap budget), 17 – Working prototype W1 (P1 geometry **Rev B** and assembly order), 15 – Bench test plan (T0 to T4) Shop numbers: `analysis/generator/manufacturing.py` → `manufacturing_summary.json`, `coil_connection.csv` Geometry: `cad/nacelle_p1/geometry.py` (single source of truth, unchanged by this document) Pictures: `cad/generator_mfg/` (`python3 figures.py` → G01 to G19); illustrated PDF in the project folder

Summary

How to use the pictures. Every work step has a picture sheet (G01 to G19) with numbered panels. Do the panels in number order. The words under each panel say what to do; the tables in the text give the exact numbers to check. Colours mean the same thing on every sheet: grey = steel, orange = copper, yellow = insulation, green = resin, light blue = aluminium, tan = glass-epoxy, **red = north magnet, blue = south magnet**. A green tick means "this is right"; a red cross means "never do this". Drawings of real parts (G01, G03, G06, G09, G12, G13, G16) are drawn to scale from the design files.

G01 What we are building: the P1 generator, cut in half

Real shape and size from geometry.py. Only the top half is drawn; the bottom half is the same.



This is the complete instruction set for making the steel P1 generator in house: the outer-rotor PM generator that replaces W1's printed dummy. It covers materials and purchasing, the stator (laminations, insulation, winding, connection, impregnation, potting), the rotor (drum, magnets, encapsulation), joining both on the spindle, and every inspection and test up to handover to the bench (doc 15).

Item	Value (from docs 06, 14, 17)
Machine	Outer-rotor radial flux PMSG, 40 poles / 45 slots, tooth-coil (concentrated) double-layer winding, star
Stator	Ø598 OD, bore Ø437.2, stack 74.0 mm, M270-35A self-bonding laminations, slot 65 mm deep, 4 mm opening
Winding	45 coils, **28 turns, 4 strands Ø1.55 mm in hand**, class 200 grade 2 wire, 15 coils per phase in series, class H, VPI + potted end windings
Rotor	S355J2+N drum, magnet seat Ø618.4, 160 + 8 spare N40UH arc segments (41 g each), 0.5 mm glass-epoxy sleeve, mechanical gap 2.0 mm
Mass	Stator ≈ 115 kg (with carrier), rotor ≈ 149 kg (with disc and covers)
Expected values	R20 = 0.217 Ω per phase; line EMF ≈ 430 V rms at 178 rpm and 20 °C (scaled, to be replaced by the FEA rerun)

One design change is needed before anything can be built (section 1): the stator as drawn cannot be wound. The 4 mm slot opening and the 50 % copper fill required in the free slot space rule out every way of putting the coils in. This document is written for the recommended fix, a **segmented stator with one tooth per segment**, where each coil is precision layer-wound directly on its own tooth and the 45 wound segments slide into dovetail grooves in the carrier. The magnetic shape (tooth, tips, slot, yoke, gap) is unchanged; the lamination and carrier drawings change. **Martin chose this route on 2026-11-03.** The other options are in Appendix C.

Build in three stages:

1. **Q1 process qualification article** (5 wound segments in a carrier sector plus magnet test panels): proves winding fit, VPI, potting, insulation and magnet bonding before any P1 parts are committed (section 10). About 6 weeks, in parallel with the magnet order.
2. **P1 stator and rotor** (sections 5 to 7).
3. **Join on the spindle and test** (sections 8 and 9), then hand over to the bench plan (doc 15).

Generator ready for the bench about **13 to 14 weeks** after the magnet order, which matches the doc 14 schedule.

1. Manufacturability review of the frozen design

Before writing instructions, every part of the doc 06/14 generator was checked for whether it can be made, inspected and repaired. Findings:

#	Finding	Consequence	Action in this document
M1	**The one-piece stator cannot be wound.** Tooth tips are 37.7 mm wide on a 22.3 mm tooth, with only a 4.0 mm slot opening. A pre-wound coil cannot pass over the tips. Needle winding through 4 mm with 4 strands of Ø1.66 mm (with enamel) is not possible. Hand drop-in, wire by wire through the opening, reaches about 40 % copper fill of the free space at best, and the design needs **49.7 %** (211 mm² of copper per coil side in 425 mm² free). Precision layer winding uses **90 %** of the 125 positions available (layers of 37, 37, 31, 17, 3 turns), which only works when the tooth is free on all sides	Doc 14 stator cannot be built as drawn	**Segmented stator** (section 5), chosen by Martin 2026-11-03 (Appendix C)
M2	Segmentation means the stack is no longer shrunk onto the carrier with 3 keys	Carrier rim gets 45 dovetail grooves; keys and the Ø437.25 shrink fit go	New drawings NP1-200 Rev C and NP1-201 Rev C (section 5.1)
M3	With segments, each sheet can no longer be rotated by 24° to average sheet thickness (doc 14)	Stack height scatter per segment	Sheets counted and stack measured per segment, ±0.1 mm (section 5.3)
M4	28 turns (doc 14) versus 27 to 28 (doc 06). Scaling doc 06 to 74 mm gives **392 V line at 100 °C with 28 turns** (target 380 V) and **≈ 920 V peak** at 250 rpm, −40 °C (doc 12 used 892 V). 27 turns gives 378 V and 889 V	Still below the 1100 V DC link capacitors and 1200 V devices, but outside the doc 12 number	Freeze the turn count from the FEA rerun before winding the 45 P1 coils; Q1 coils use 28 (section 5.5)
M5	A hot-cured magnet adhesive (150 °C) locks in shear strain: steel shrinks 1.9×10^{-3} more than NdFeB from 120 °C to −40 °C, which is about 0.07 mm over a 38 mm magnet on a 0.15 mm bond line	Bond cracking in cold cycling	**Room-temperature-curing toughened epoxy, post-cure at 80 °C** (section 6.5). In an outer rotor the magnet pull and centrifugal force both push the magnets into the drum, so the bond only carries torque shear (≈ 20 kPa) and must not crack
M6	The 0.5 mm glass-epoxy cover layer, laid by hand, carries the full magnet thickness and bond-line scatter into the air gap (0.05 mm in the budget)	Gap scatter	**Mould the sleeve against a precision non-magnetic mandrel** located on the bearing seats (section 6.7): the sleeve bore becomes a controlled Ø602.0 surface concentric with the bearings
M7	Magnets are arcs, so a piece cannot be flipped to change polarity	Two different parts must be ordered	**NP1-220-N** (north on the concave face) and **NP1-220-S**, 84 of each (section 4.3)
M8	Magnet axial segmentation (4 per pole) still waits for the eddy-loss check (doc 14 open point 3)	Order cannot be placed with confidence	Run the eddy-loss check in week 1; if it changes the count, only the magnet drawing changes

#	Finding	Consequence	Action in this document
M9	A shrink-fitted one-piece stack could not be repaired; a damaged coil meant a new stator	Service cost over 20 years	Segments can be pulled and replaced **before VPI**; after VPI a damaged coil means re-winding that segment in a stripped stator (section 11)

M1 is the only show-stopper. M2, M3, M6 and M7 follow from the chosen fixes; M4, M5 and M8 are decisions or checks with a clear owner.

2. Facilities, equipment and people

2.1 Work areas

Area	Purpose	Requirements
A. Machine shop	Drum, carrier, rings, tools	Boring mill or vertical lathe (Ø900 swing), CNC mill, wire EDM (carrier grooves), grinding
B. Lamination cell	Cut, deburr, bond segments	Fibre laser with N_2 assist (P1) or progressive die (series), bonding press with heated platens, fixture NP1-T03
C. Winding room	Insulate and wind segments, assemble and connect stator	Clean (no metal grinding nearby), 20 to 25 °C, RH ≤ 60 %, ESD-safe benches, programmable spindle winder with 4-strand tensioner, brazing station
D. Impregnation bay	VPI, ovens, potting	VPI tank ≥ Ø900 × 600 mm (vacuum ≤ 1 mbar, pressure 6 bar), curing oven ≥ 180 °C with rotisserie, resin storage 5 to 25 °C, extraction ventilation
E. Magnet room	Magnet storage, flux check, bonding, encapsulation	**No loose steel**, non-magnetic benches (aluminium, wood, composite), floor marking of the 0.5 mT line, pacemaker warning on the door, access list, non-magnetic rotary table
F. Test lab	Electrical tests	Fenced HV test area to EN 50191 with interlocks and warning lamps, hipot, surge tester, PD detector, micro-ohmmeter, insulation tester, LCR meter
G. Assembly bay	Joining stator and rotor on the spindle	Overhead crane ≥ 1 t with textile slings, vertical assembly stand with tool NP1-T01, magnetic-hazard zone marking

2.2 Tools and fixtures

No.	Tool	Purpose
NP1-T01	Drum assembly guide (exists in doc 14)	Lowers the magnetised rotor over the stator on the spindle with a screw jack
NP1-T02	Magnet insertion jig (doc 14, detailed in section 6.5)	Slides each magnet into its channel and lowers it under screw control
NP1-T03	Segment bonding fixture (Rev C: for segments)	Stacks and bonds 205 sheets per segment on a precision dovetail and tip gauge
NP1-T04	Segment winding chuck	Holds a segment on the winder spindle by its dovetail, with end-flange guides
NP1-T05	Stator assembly and measuring stand	Carrier on a rotary table with a probe on the tooth tips; inserts segments axially
NP1-T06	Sleeve moulding mandrel	Aluminium, Ø602.0, located on both cup seats; moulds the magnet sleeve
NP1-T07	VPI frame	Lifts and turns the stator in the tank and oven
NP1-T08	Potting moulds	Silicone moulds for front and rear end windings
NP1-T09	Magnet flux tester	Helmholtz coil and fluxmeter, non-magnetic holder
NP1-T10	Pole scanner	Hall probe on an arm over the non-magnetic rotary table, maps the field of every pole
NP1-T11	Non-magnetic hand tools	Titanium, beryllium-copper or austenitic stainless (check with a magnet) tools, brass and plastic shims, G10 gap feelers 1.5 mm

2.3 People

- **Winding and connection:** trained winder; brazing to EN ISO 13585.
- **Magnet work:** at least two trained people, one acting only as safety watch. Written procedure signed before each session.
- **HV testing:** electrically skilled person to EN 50110 / EN 50191.
- **Welding:** EN ISO 9606-1 welders, EN ISO 3834-3 quality level, NDT level 2 to EN ISO 9712.

2.4 Safety rules (all generator work)

G19 Safety: read this before any magnet, resin or high-voltage work



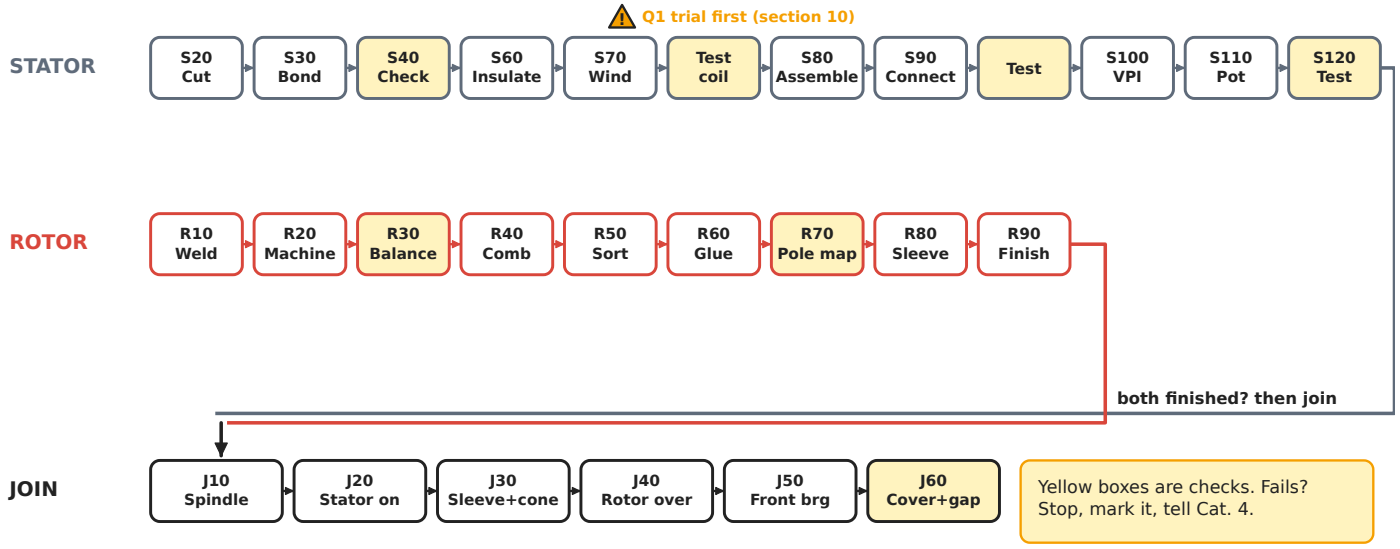
Hazard	Rule
Magnet crushing and pinching	One magnet piece pulls ≈ 280 N (≈ 28 kg) on thick steel, more on another magnet. Never carry two loose pieces in one hand; keep pieces ≥ 300 mm apart unless in spacers; handle with the jig only near the drum; no rings or watches; cut-resistant gloves and safety glasses (NdFeB chips like glass)
Assembled rotor	≈ 27 kN total radial pull on the stator, rising by **1.9 kN per mm** off centre. Only the guided, screw-controlled procedure in section 8 is allowed. Never put hands in the drum when the stator is near it
Medical implants	Nobody with a pacemaker or ICD in the magnet room or assembly bay; signs at every entrance
Magnetic media and instruments	Keep phones, cards, watches and instruments outside the 0.5 mT line
Magnet dust	No grinding or machining of magnets. NdFeB dust is pyrophoric. Broken pieces go into a sealed steel tin under oil, then to the supplier for recycling
HV tests	Tests only inside the fenced area, two-hand or key interlock, discharge and earth every winding after every test (5 × the test time, then earth stick)
Resins and solvents	Epoxies sensitise skin: nitrile gloves, sleeves, eye protection, local extraction; read the SDS before first use; no food in areas C, D, E
Hot work	Ovens up to 180 °C, brazing: heat gloves, fire watch, extinguisher class B and electrical
Lifting	Textile slings only near the magnetised rotor. Drum with magnets ≈ 105 kg, complete rotor ≈ 150 kg, stator ≈ 115 kg

3. Process route and schedule

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Week:      0    1    2    3    4    5    6    7    8    9   10   11   12   13   14
Decisions  M1/M4/M8
FEA rerun  ■■■
Drawings   ■■■■■■■■ (NP1-200/201 Rev C, T03 to T10)
Magnets         ordered ──────────── 8 to 12 weeks ──────────► in
Q1 article           ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ (section 10)
Laminations               ■■■■■■■■■■■■ cut, bond, inspect
Carrier, drum        ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ weld, PWHT, machine
Winding                                      ■■■■■■■ 45 segments
Stator assy, VPI, pot                           ■■■■■■■
Rotor magnets, sleeve                              ■■■■■■■
Join on spindle, tests                                ■■■■■■
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G02 The order of work: two tracks side by side, then they meet

Each box is one operation in section 3. Do them left to right. Do not skip a test box.



Op	Stator track	Op	Rotor track
S10	Receive and inspect steel coil, wire, insulation, resin	R10	Drum weldment, NDT, stress relief
S20	Cut segment laminations	R20	Machine drum in one set-up
S30	Bond segment stacks (45 + 3 spare)	R30	Pre-balance bare drum (G6.3)
S40	Inspect segments	R40	Prepare magnet seat, bond pole comb
S50	Machine carrier, dovetail grooves	R50	Receive, flux test and sort magnets
S60	Insulate segments	R60	Bond magnets (N poles, then S poles)
S70	Wind coils, test each coil	R70	Pole map
S80	Assemble segments on carrier, wedges, sensors	R80	Mould the sleeve with mandrel T06
S90	Connect, pre-VPI tests	R90	Rotor inspection, cups, rear seal cover, disc
S100	VPI and cure		
S110	Pot end windings		
S120	Final stator tests		

Op	Stator track	Op	Rotor track
	Join	J10 to J60	Section 8; then tests section 9

4. Materials and purchasing specifications

Every lot needs a 3.1 inspection certificate (EN 10204) unless noted. Incoming inspection results go into the generator record (section 11).

4.1 Electrical steel and stator materials

Item	Specification	Quantity	Incoming check
Lamination steel	**M270-35A** (EN 10106), 0.35 mm, with **self-bonding varnish** both sides (e.g. "backlack" type), C5/C6 insulation	9,225 segment sheets (205 × 45) + 5 % + Q1 → order ≈ **0.6 t** of slit coil	Thickness 10 points ±0.02, varnish thickness, supplier Epstein loss ≤ 2.70 W/kg at 1.5 T 50 Hz, bonding test coupon
Winding wire	Enamelled round copper **Ø1.55 mm, grade 2, thermal class 200** (polyesterimide + polyamide-imide overcoat), IEC 60317-13; partial-discharge-resistant (corona-resistant) enamel preferred because of converter feed	**25 kg** (20.1 kg in the winding + 25 %), one lot	Diameter bare and overall (max 1.66), elongation ≥ 30 %, breakdown voltage per IEC 60851-5, pinholes
Slot liner	Nomex-Kapton-Nomex (NKN), 0.30 mm, class H/200	5 m²	Thickness, breakdown
Phase separators	NKN 0.25 mm (or 0.5 mm aramid paper)	2 m²	
Slot wedges	Glass-epoxy (G11, class H), profile to suit the wedge seat, 2.0 mm	45 × 74 mm + spares	Profile gauge
End caps	Moulded **PPS GF40** (or PEI), class H, one front and one rear per segment	96	Fit on master segment
Lead wire	Silicone-insulated tinned flexible copper, 10 mm², 180 °C, 1 kV (or EPR class H)	15 m	
Sleeving and ties	Glass/silicone sleeving class H, polyester-glass lacing cord, glass tape	Kit	
Brazing	BCuP-2 (CuP7) rods (flux-free on copper), copper crimp ferrules	Kit	
VPI resin	Solvent-free **class H epoxy** VPI resin (single component or two-component with long pot life), approved by the wire maker for the enamel; tan δ, viscosity and gel time data	1 tank fill + 20 kg uptake reserve	Viscosity and gel time on arrival and before each VPI
Potting compound	Two-component **filled epoxy**, class H, thermal conductivity ≥ 0.8 W/m K, CTE ≤ 35 ppm/K, low exotherm, toughened for −40 °C cycling	≈ 8 kg + 3 kg Q1	Mixed sample hardness, gel time
Temperature sensors	Thin-film **PT100 class A**, 3-wire, Kapton/glass sleeved, fit 2 mm × 4 mm in the separator gap	8 (6 used, 2 spare) + 4 for Q1	Resistance at 0 °C ±0.15 Ω
Carrier steel	S355J2+N ring forging or plate, normalised	1 (≈ 70 kg raw)	Certificate, UT of forging

4.2 Rotor materials

Item	Specification	Quantity
Drum shell	**S355J2+N seamless ring-rolled forging** preferred (no longitudinal weld through the magnet yoke); plate-rolled with full-penetration seam acceptable for P1 if 100 % UT	1

Item	Specification	Quantity
Drum webs, flange ring	S355J2+N plate 8, 20 mm; bearing tube S355J2H	per drawing NP1-210
Pole comb bars	GRP pultruded bar (glass-epoxy, class H), 9.6 × 7.0 mm, 74 mm long; or machined from G11 sheet	40 + 5
Magnet end rings	G11 rings, front and rear, closing the magnet pack ends	2
Magnet adhesive	Two-component **toughened epoxy, room-temperature cure**, service −40 to +150 °C, Tg ≥ 120 °C after an 80 °C post-cure, lap shear ≥ 15 MPa (23 °C) and ≥ 5 MPa (120 °C) on NiCuNi-epoxy coated NdFeB to grit-blasted steel, with **0.15 mm glass beads** for bond line, approved by the magnet maker for its coating	1.5 kg incl. Q1
Sleeve fabric	E-glass twill 2 × 100 g/m² (≈ 0.2 mm cured total, layup tuned to 0.4 to 0.6 mm) + glass-scrim peel ply	3 m²
Sleeve resin	Low-viscosity infusion epoxy, RT cure, post-cure 80 °C, Tg ≥ 100 °C	2 kg
Primer	Adhesive maker's primer for steel	

4.3 Magnets NP1-220 (purchase specification)

Item	Requirement
Material	Sintered NdFeB **N40UH** (Br ≥ 1.26 T, HcJ ≥ 1,990 kA/m at 20 °C); batch demagnetisation curves (J-H and B-H) at 20, 60, 100, 120 and 150 °C; temperature coefficients of Br and HcJ stated
Shape	Radial arc segment, **R301.5 concave / R309.2 convex, arc 7.20°** (chord 37.86 inside, 38.83 outside), **axial length 18.35**, thickness 7.70
Tolerances	Thickness ±0.03, arc radii ±0.05, length ±0.05, arc angle ±0.05°; edges chamfered 0.2 to 0.3
Magnetisation	**Radial**, fully saturated. Two variants: **NP1-220-N** (north pole on the concave, stator-facing side) and **NP1-220-S**; each piece marked with a coloured dot on the convex face (red = N inside, blue = S inside)
Quantity	84 N + 84 S (80 + 4 spare each), plus 8 test pieces (unmagnetised acceptable) for Q1 bond and salt-mist tests
Coating	**NiCuNi** (≥ 15 µm) plus **epoxy top coat** (≥ 15 µm); ≥ 500 h neutral salt spray (ISO 9227) without red rust on test pieces; adhesion approved by the adhesive maker
Uniformity	Magnetic moment of every piece measured by the supplier, within **±1.5 %** of the lot mean; data per piece serial
Packaging	Individually in non-magnetic spacers, repelling layout, steel-shielded box for air freight (IATA PI 953 field limit)
Order note	**Longest lead item, order first** (8 to 12 weeks), after the eddy-loss check (M8)

4.4 Fasteners and bought parts

As doc 14 BOM: bearings 32028 X, KMT 27, seals, M12 10.9 carrier screws with dowels, M10 10.9 disc screws, M16 10.9 hub screws. All fasteners for the generator interior **A4-80 stainless** where they sit within 100 mm of a magnet face (non-magnetic during assembly); 10.9 steel elsewhere.

5. Stator

5.1 Segment and carrier design changes (drawing basis for Rev C)

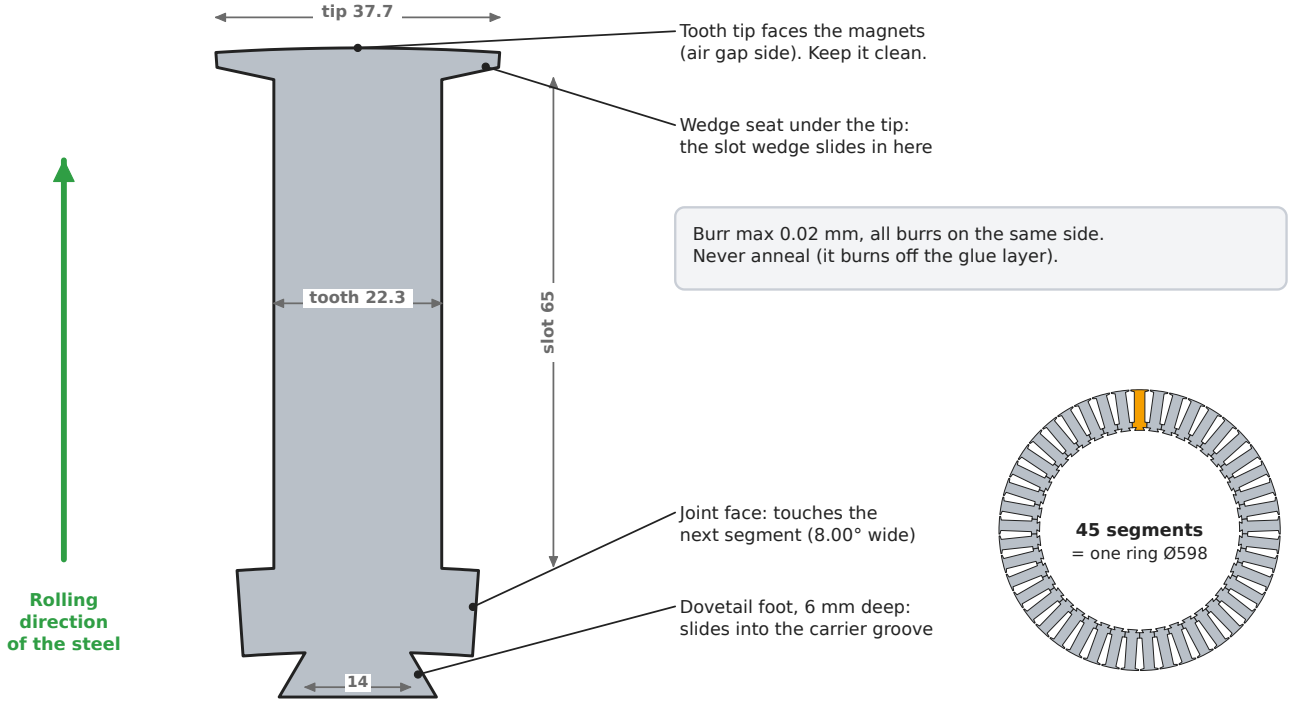
Feature	Rev C definition
Segment NP1-201	One tooth (22.3 mm) with tips, plus one slot pitch of yoke (**8.00°**), joints on the slot centre lines in the yoke. Magnetic contour inside the joints is identical to `lamination_outline()` in geometry.py
Dovetail foot	Below the yoke, r 218.6 → 212.6 (6 mm deep), 14 mm wide at r 218.6, 60° flanks; the carrier rim OD stays Ø437.2
Carrier NP1-200	Rim OD Ø437.2 with **45 dovetail grooves**, angular position ±0.01°, radial seat ±0.01 to the bolting face axis; grooves open to the front (x 343.5), closed by the rear stack stop (x 417). The 3 keyways and the shrink fit are removed
Joint fit	Foot in groove: 0.01 to 0.03 mm clearance (slides by hand when clean), fixed by VPI resin and clamped axially by NP1-202
Tip seats	Tooth-tip underside carries the 2 mm wedge seat (as Rev B)
Air gap budget	Item "stack OD to bore 0.050" becomes "segment tip radius in assembly 0.050": segment tip radius ±0.02, groove seat ±0.01, plus foot clearance taken up by VPI. Total budget unchanged
FEA rerun	Model the 45 yoke joints (0.02 mm effective gap each) and a ±0.02 mm random radial scatter of segments; confirm EMF, cogging ≤ 0.5 % and torque ripple. Expected EMF loss < 1 %

Spare segments: 3, wound and tested. All segments are identical, so any spare replaces any tooth.

5.2 S20 Cut laminations

G03 One segment sheet (NP1-201 Rev C): 0.35 mm thin, cut by laser

Drawn to scale from the frozen design. 205 sheets glued together make one segment; 45 segments make the stator.



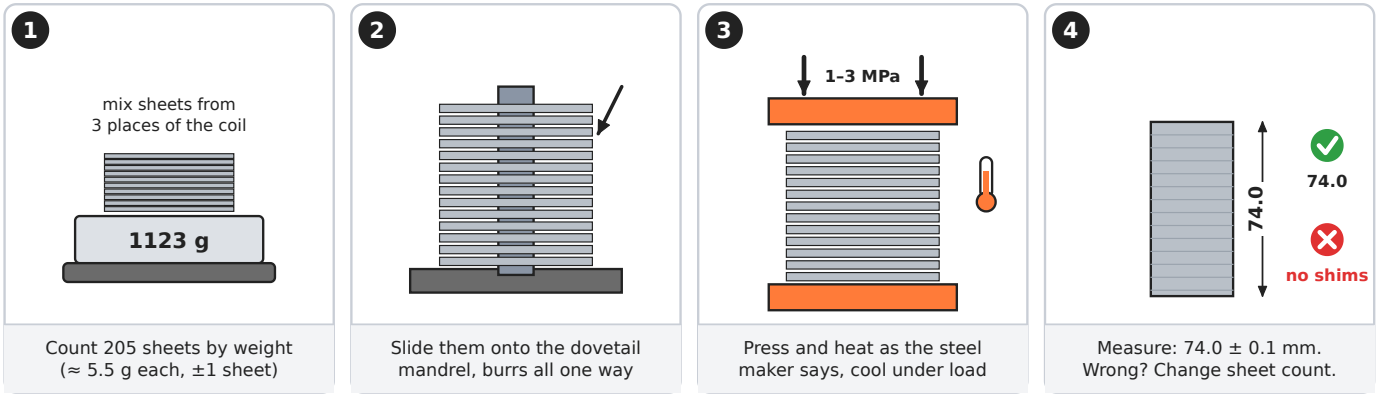
1. Check the coil certificate and incoming tests (4.1). Mark the rolling direction on the coil. Segments are cut with the **tooth axis along the rolling direction** (lower loss in the tooth,

where flux density is highest).

2. **P1 (laser):** fibre laser, nitrogen assist, parameters set on test sheets for minimum heat input; cut contour from the Rev C DXF. Burr height ≤ **0.02 mm**, measured on 1 in 50 sheets. Do not anneal (it destroys the bonding varnish); the laser edge damage is covered by the 1.5 loss build factor in doc 05.
3. **Series (later):** progressive die, burr ≤ 0.02 mm, die resharpened by burr height.
4. Every 200th sheet: measure tooth width, tip width, dovetail profile with the optical comparator or CMM; tolerance ±0.02 mm.
5. Keep sheets flat, dry and free of oil. Stack in bundles of 205 counted by weight (≈ 5.5 g per sheet, ≈ 1.12 kg per bundle; weigh the bundle, ±1 sheet).

5.3 S30 Bond segment stacks (fixture NP1-T03)

G04 Stack and bond one segment (fixture NP1-T03)



Count 205 sheets by weight (≈ 5.5 g each, ±1 sheet) | Slide them onto the dovetail mandrel, burrs all one way | Press and heat as the steel maker says, cool under load | Measure: 74.0 ± 0.1 mm. Wrong? Change sheet count.

1. Clean the fixture: dovetail mandrel, tip locator, release-coated platens.
2. Load **205 sheets**, burrs all in the same direction. Mix sheets from at least three positions across the coil width in each segment to average thickness (replaces the doc 14 rotation, M3).
3. Close the fixture; apply the varnish maker's bonding cycle (typically 1 to 3 MPa at 150 to 220 °C for a set time). Record pressure, temperature and time.
4. Cool under pressure to below 60 °C, then release.
5. **Stack height 74.0 ±0.1 mm**. If out, add or remove sheets and re-bond the next segment; never shim.
6. Mark the segment serial on the yoke back face (laser or vibro-etch, not stamping).

5.4 S40 Inspect segments

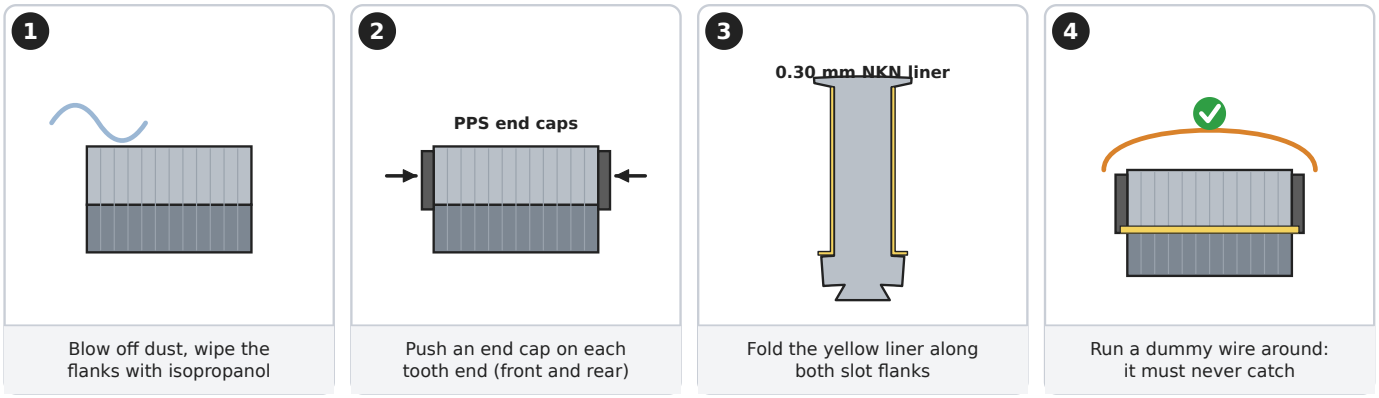
Check	Method	Accept
Height	Micrometer at 4 points	74.0 ±0.1
Squareness of stack to dovetail	Square and feelers	≤ 0.05 over 74
Tip radius from dovetail datum	Fixture gauge with dial	±0.02 (key item of the air gap budget)
Tooth width, slot flanks	Go/no-go gauge	±0.03
Bond strength	Peel by hand at one end corner (no delamination); 1 sacrificial segment per batch in a tensile test	≥ varnish datasheet
Interlaminar insulation	4-wire resistance top-to-bottom of the stack at 1 A DC	≥ 10 Ω (no through-shorts by burrs)

Check	Method	Accept
Burrs, steps, loose sheets	Visual, 10× loupe	None at slot flanks and tips

Make **48 segments** (45 + 3 spare).

5.5 S60 Insulate segments

G05 Dress the segment before winding: end caps and slot liner



1 — Blow off dust, wipe the flanks with isopropanol

2 — Push an end cap on each tooth end (front and rear)

3 — Fold the yellow liner along both slot flanks

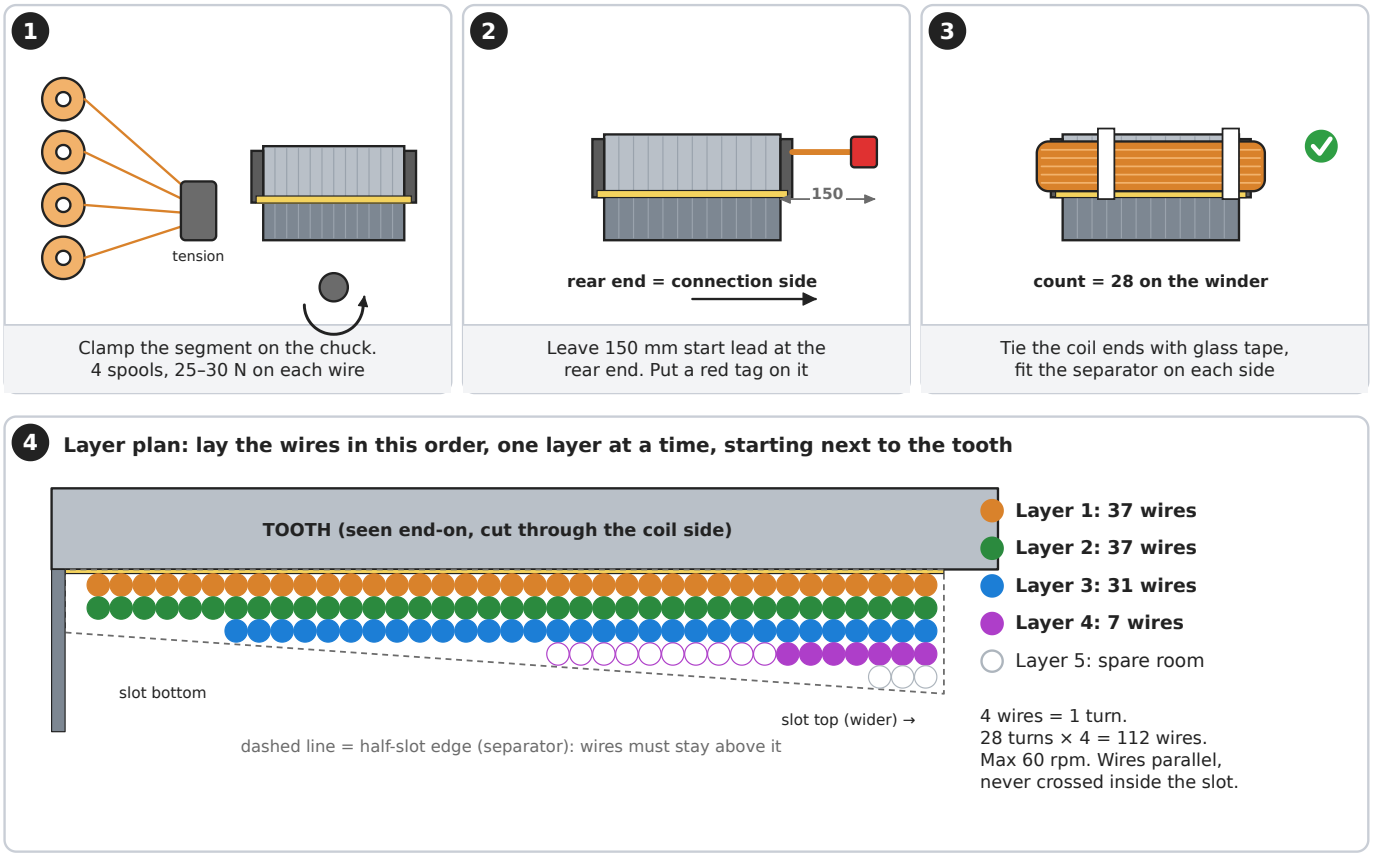
4 — Run a dummy wire around: it must never catch

1. Blow off with dry filtered air, wipe flanks with isopropanol.
2. Fit **front and rear end caps** (PPS GF40) onto the tooth ends; they round the coil corners (R ≥ 3 mm) and carry the coil 3 mm clear of the stack ends.
3. Fold the **NKN 0.30 slot liner** to the slot flank shape (yoke face, tooth flank, under the tip) on both sides of the tooth. Liner cuffs 5 mm beyond each stack face, folded over the end caps. No liner may cover the dovetail or the tip outer surface.
4. Check: no sharp edge reaches wire anywhere; a 1.7 mm dummy wire run around the tooth must not catch.

5.6 S70 Wind coils (chuck NP1-T04)

G06 Wind one coil: 28 turns, 4 wires side by side (chuck NP1-T04)

Every coil is wound the same way. Reversing happens later, when connecting (G09).



Turn count: 28 for Q1 and P1 unless the FEA rerun (M4) freezes 27. The count is set in the winder program and signed off on the traveller.

1. Mount the insulated segment on the chuck by its dovetail. The chuck end flanges follow the slot flank and the slot centreline, so the coil cannot grow into the neighbour's half-slot (half free width 0.25 mm under the separator).

2. Load **4 strands Ø1.55 mm** from four spools of the same wire lot through the 4-strand tensioner. Tension **25 to 30 N per strand** (≈ 13 to 16 MPa, far below the yield of annealed copper). Strands must lie parallel and must not cross inside the slot part of the coil.

3. Leave the **start lead 150 mm** long at the rear end (connection side), at the slot bottom.

4. Wind **layer by layer**, starting at the tooth flank, outward. The layer plan (each turn = 4 strand positions, positions counted along the tooth height):

Layer (from tooth flank)	Strand positions available	Strand positions used
1	37	37
2	37	37
3	31	31
4	17	7
5	3	0 (reserve)

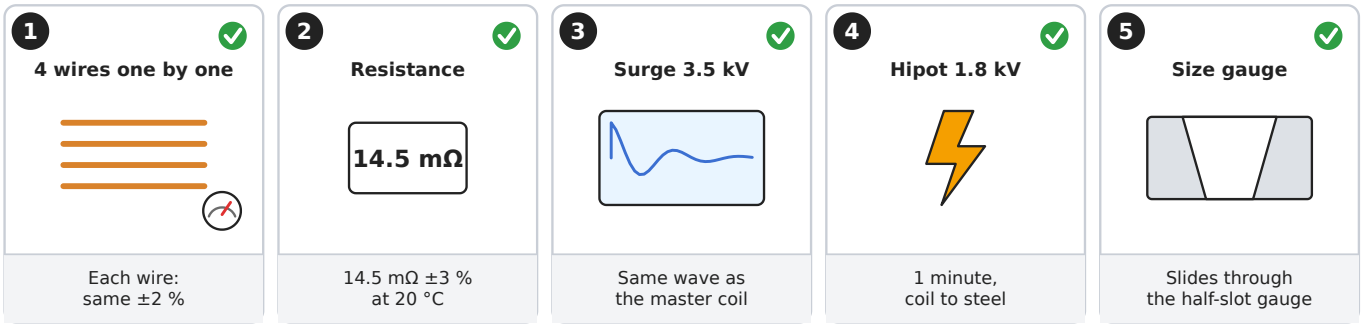
(28 turns × 4 strands = 112 positions needed, 125 available.) Layers step outward toward the slot top, where the slot is wider; the program steps the traverse limit per layer. Speed ≤ 60 rpm.

5. Finish lead **150 mm** at the rear end. Total wire per coil ≈ **6.6 m per strand**.
6. Tie the coil end turns with one turn of glass tape at each end cap; tape the leads with glass sleeving, mark the **start lead** with a red tag.
7. Fit the **phase separator** (NKN 0.25) on each coil side face; it is the boundary to the neighbouring segment's coil.

Every coil is wound in the same direction. Reversal of "−" coils is done at connection (5.9), so all 48 segments are interchangeable.

5.7 Test every coil (before assembly)

G07 Test every coil before it goes into the stator (all 48)



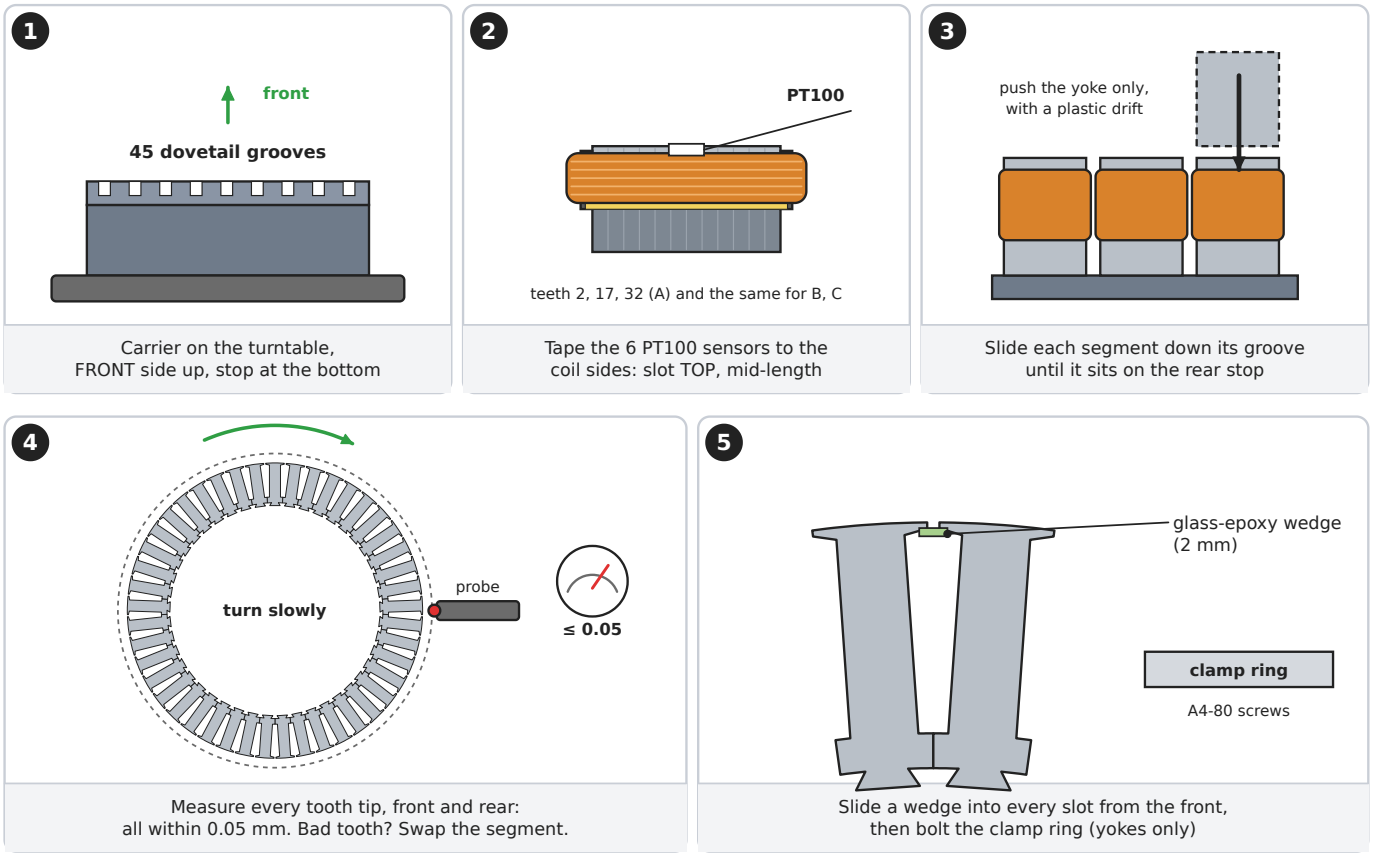
Fail: unwind, new liner, wind again on the same segment.

Test	How	Accept
Strand continuity	Each of the 4 strands separately, micro-ohmmeter	Strand resistances within ±2 % of each other (finds a broken or nicked strand)
Coil resistance	4 strands joined at each end, 4-wire, corrected to 20 °C	**14.5 mΩ ±3 %** (calculated, confirm on Q1), and within ±1 % of the running mean
Turns and inter-turn insulation	Surge comparison against the master coil, **3.5 kV peak**, rise time 0.1 to 0.2 µs	Waveform matches master (error area ratio ≤ 5 %); no collapse
Coil to core	AC hipot **1.8 kV, 1 min** (80 % of the final test, because not yet impregnated)	No breakdown, leakage stable
Dimensions	Coil side envelope gauge (half-slot profile)	Passes freely along the full length

Record results by segment serial. Failed coils are unwound and rewound on the same segment after replacing the liner.

5.8 S80 Assemble segments onto the carrier (stand NP1-T05)

G08 Put the 45 wound segments onto the carrier (stand NP1-T05)



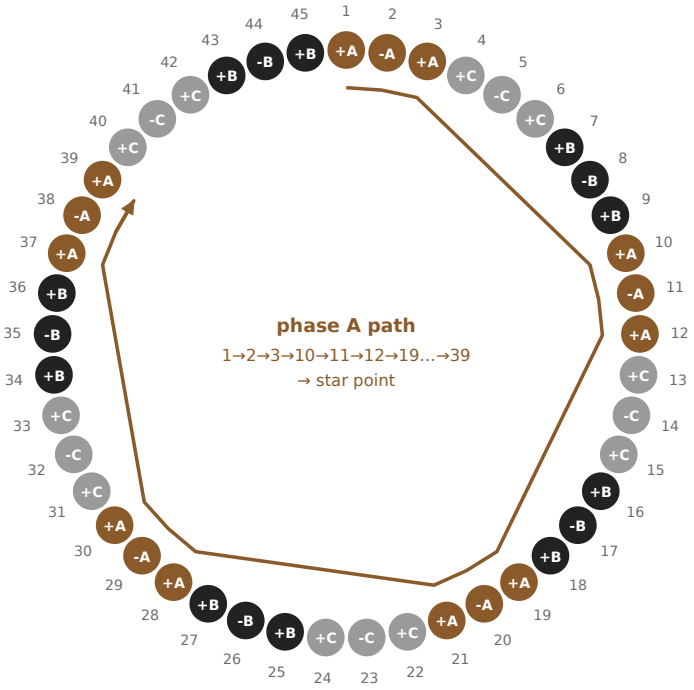
Prerequisite: carrier NP1-200 Rev C finish-machined, grooves deburred and measured (groove position ±0.01°, seat radius ±0.01), cleaned, bolting face protected.

1. Carrier horizontal on the rotary table, front end (x 343.5 side) up, rear stack stop down.
2. Fit the **6 PT100 sensors** first to the coil sides that will sit at the hottest positions: **slot top (under the wedge), mid-stack**, two per phase, spread around the circumference (suggested teeth 2, 17, 32 for phase A, and the same pattern shifted to B and C). Sensor leads run with the coil leads to the rear.
3. Slide segments axially into their grooves **in tooth order 1 to 45** (Appendix A). The coil sides pass the neighbour's coil side with the separator between; push with a plastic drift only on the yoke, never on the coil. Each segment must seat against the rear stop.
4. After all 45 are in, measure **tooth-tip radius at every tooth**, front and rear end, with the probe on the stand: run-out **≤ 0.05 mm** TIR. A tooth out of tolerance: change the segment for a spare and recheck.
5. Insert the **slot wedges** axially from the front in every slot.
6. Fit the front clamp ring NP1-202 (A4-80 screws, torque per drawing). It clamps the yokes only, not the coils.
7. Check: no coil touches the clamp ring, the carrier web, or the rear stack stop; the end windings stay within **x 321 to 439** and **r ≤ 296**.

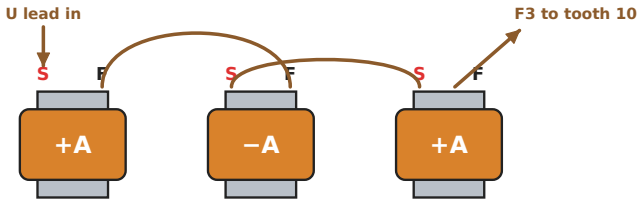
5.9 S90 Connect the winding

G09 Connect the coils: each phase = 15 coils in a row

Look at the stator from the REAR (connection side). Tooth 1 is at 12 o'clock at the carrier mark.



How to join one group (teeth 1, 2, 3):



S = start (red tag), F = finish.
+ coil: current goes in at S. − coil: in at F.
So: F1 joins F2, S2 joins S3, F3 goes on to tooth 10.

Phase A = U (brown), starts at tooth 1
Phase B = V (black), starts at tooth 7
Phase C = W (grey), starts at tooth 4

Long jumpers between groups run on the glass-epoxy connection ring, one track per phase. The 3 phase ends meet in the star point. Then check polarity with a compass (5.10).

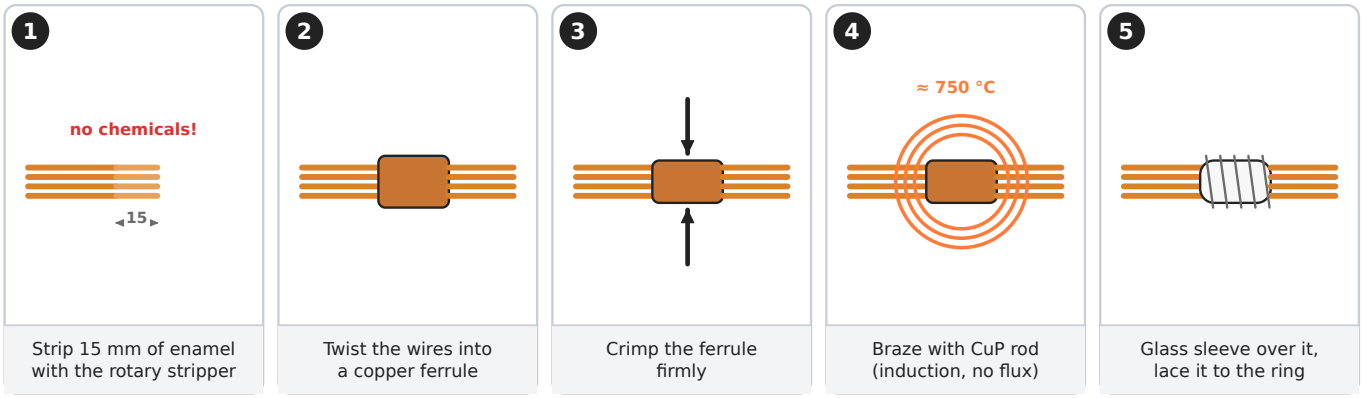
Connection plan: coil_connection.csv, summarised in Appendix A. Each phase has five groups of three adjacent coils, **(+, −, +)**, with the groups nine teeth apart. Phases around the stator: **A, C, B**, A, C, B …

- A "+" coil is connected start → finish in the phase's current direction; a "−" coil finish → start.
- Within a group the jumpers are short (adjacent teeth). Between groups the jumpers run on a **connection ring** of glass-epoxy at the rear end winding, one track per phase.

For each joint:

1. Strip the enamel mechanically (rotary stripper) or by laser over 15 mm; **no chemical strippers** (residues attack the VPI resin).
2. Twist the 4 + 4 strands (or the 4 strands plus the jumper) into a copper **crimp ferrule**, crimp, then **induction-braze with BCuP-2** (flux-free on copper). Joint resistance ≤ that of 20 mm of the coil conductor.
3. Clean, inspect for full fill and no sharp ends, sleeve with class H glass sleeving, lace to the connection ring.
4. **Star point**: the three phase ends brazed into one ferrule, insulated and fixed on the ring.
5. **Phase leads** U, V, W: brazed to 10 mm² silicone leads, out through the two Ø28 holes in the carrier web and spindle flange (doc 14), with a strain relief tie on the ring. Tinned lugs at the terminal box (H_2S, doc 04).
6. PT100 leads: twisted triples, out through the same holes in a separate sleeve.

G10 Make one brazed joint (same for every joint)



1	2	3	4	5
Strip 15 mm of enamel with the rotary stripper	Twist the wires into a copper ferrule	Crimp the ferrule firmly	Braze with CuP rod (induction, no flux)	Glass sleeve over it, lace it to the ring

5.10 Pre-VPI tests

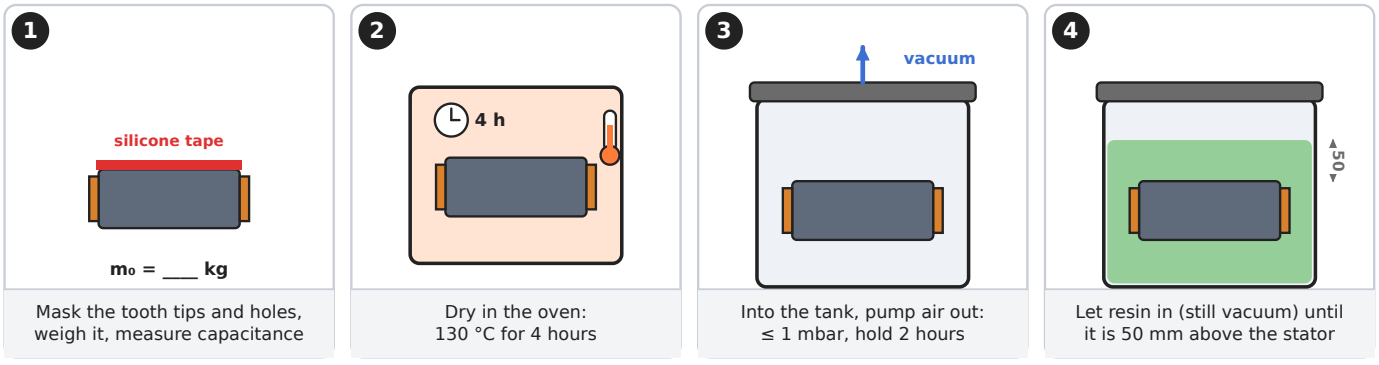
Test	Accept
Phase resistance at 20 °C	**0.217 Ω** ±3 % per phase; the three phases within **±1 %** of each other
Insulation resistance, 1 kV DC, 1 min	≥ 500 MΩ (dry, unimpregnated)
Surge, phase against phase, 3.5 kV peak	Waveforms match within 5 % (finds a reversed or shorted coil)
Polarity check	Low-voltage DC (2 A) into each phase, compass or Hall probe at each tooth tip: pattern **N S N, S N S** … matches Appendix A
PT100	Each sensor reads ambient ±0.5 K

This is the last point where a segment can be changed easily. Do not start VPI until all pre-VPI tests pass.

5.11 S100 Vacuum-pressure impregnation (frame NP1-T07)

G11 Soak the stator in resin under vacuum and pressure (VPI)

The resin glues every wire, fills every gap and keeps water out for 20 years.



One VPI run at a glance

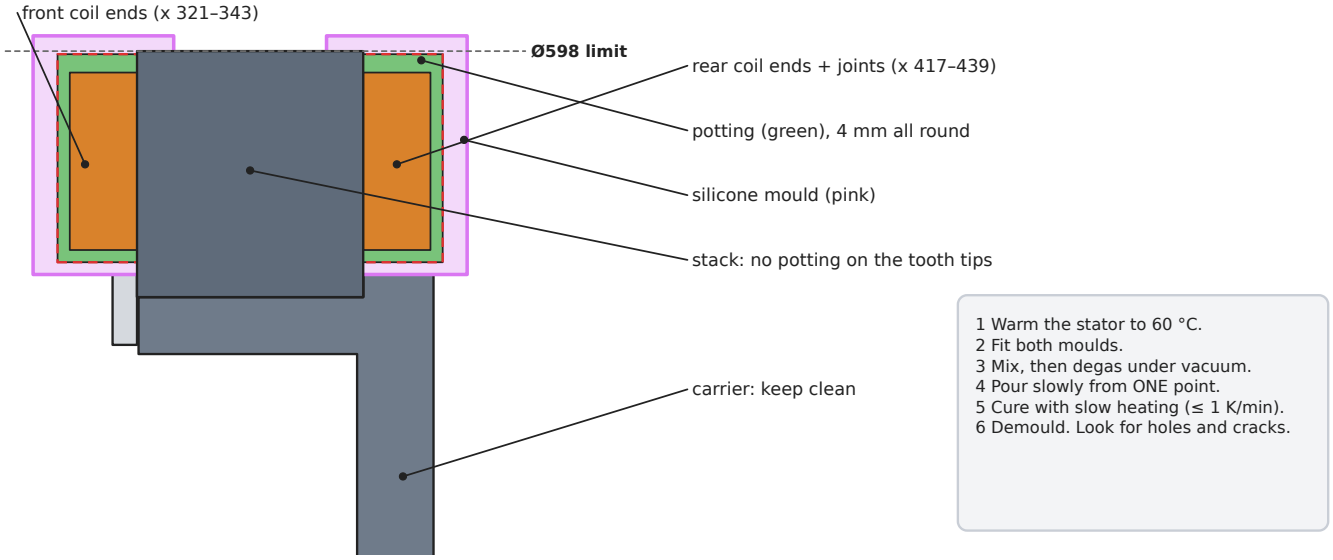


1. **Mask** the tooth-tip OD (air gap surface), the carrier bolting face, dowel holes, bolt holes, the Ø28 cable holes and lead ends with silicone tape and plugs.
2. Weigh the stator (record m_0). Measure **capacitance and tan δ** phase to core at 1 kHz (record).
3. **Pre-dry** in the oven: 4 h at 130 °C (removes moisture from liners and the winding).
4. Cool to resin temperature (typically 50 to 60 °C) and place in the tank.
5. Vacuum to **≤ 1 mbar** and hold **2 h** (hold until the pressure rise test with the pump isolated is < 0.5 mbar in 5 min).
6. Admit resin under vacuum until the stator is covered by **≥ 50 mm**; hold under vacuum 15 min.
7. Pressurise with **dry nitrogen to 5 bar**, hold **2 h**.
8. Release, lift, drain 30 min while turning the frame slowly.
9. **Cure rotating** in the oven per the resin datasheet (typically 6 to 10 h at 150 to 160 °C), turning at 2 to 5 rpm until gelled, so resin does not drain from the top slots.
10. Remove masking; clean the tooth tips; no resin build-up on the tip OD (max 0.05 mm, check with the stand probe; remove by hand with a plastic scraper, never by grinding).
11. Weigh again (m_1): resin uptake $\mathbf{m_1 - m_0 \approx 3}$ **to 5 kg** (target set on Q1). Measure capacitance and tan δ again: capacitance must rise (Q1 sets the minimum rise that showed full impregnation in the section cut).

5.12 S110 Pot the end windings (moulds NP1-T08)

G12 Pour potting around the coil ends (moulds NP1-T08)

Section through the stator, real size scale. Green = where the potting goes. It must stay inside the dashed lines.



1. Pre-heat the stator to 60 °C.
2. Fit the front and rear silicone moulds; they define the potting envelope: **x 321 to 343 and x 417 to 439, r 232 to 296** (stays inside the stator OD, away from the clamp ring bolts and the carrier web).
3. Mix the filled epoxy, **degas** under vacuum, pour slowly from one point so air leaves ahead of the front; use a vacuum chamber for the stator if available.
4. Cure per datasheet with a slow ramp (≤ 1 K/min) to keep exotherm and shrinkage stress low. Demould, deburr, check for voids and cracks.
5. Protect the connection ring joints: they are inside the rear potting.

5.13 S120 Final stator tests (doc 15, T0)

All at 20 ±5 °C, results corrected to 20 °C and recorded.

Test	Standard	Accept
Phase resistance	4-wire	0.217 Ω ±3 %, balance ±1 % (unchanged from pre-VPI)
Insulation resistance, 1 kV DC	IEEE 43	**≥ 1,000 MΩ** at 40 °C equivalent; **PI ≥ 2.0**
AC hipot, phases together to core	IEC 60034-1	**2.3 kV rms, 1 min** (2 U + 1 kV with U = 630 V, the highest rms line voltage at 250 rpm cold). Any repeat test at 80 % (1.8 kV)
Hipot, phase to phase	IEC 60034-1	2.3 kV rms, 1 min, star point opened if accessible; otherwise surge only
Surge, each phase	IEC 60034-15 method	**3.5 kV peak**, phases match within 5 %
Partial discharge inception	IEC 60034-18-41 (type I)	**PDIV ≥ 1.9 kV peak** phase to core (1.2 kV winding peak from doc 12 × 1.25 ageing × 1.25 temperature/humidity; confirm the factors in the Q1 qualification)
Capacitance and tan δ	1 kHz	Within the Q1 band
Tooth-tip run-out	Stand NP1-T05	**≤ 0.05 TIR**
PT100		All 6 + spares read ambient ±0.5 K
Visual		No cracks in potting, no resin on gap or bolting surfaces

Test	Standard	Accept
Mass		Record (≈ 115 kg with carrier)

The stator is then stored covered, with a desiccant bag, until joining.

6. Rotor

6.1 R10 Drum weldment NP1-210

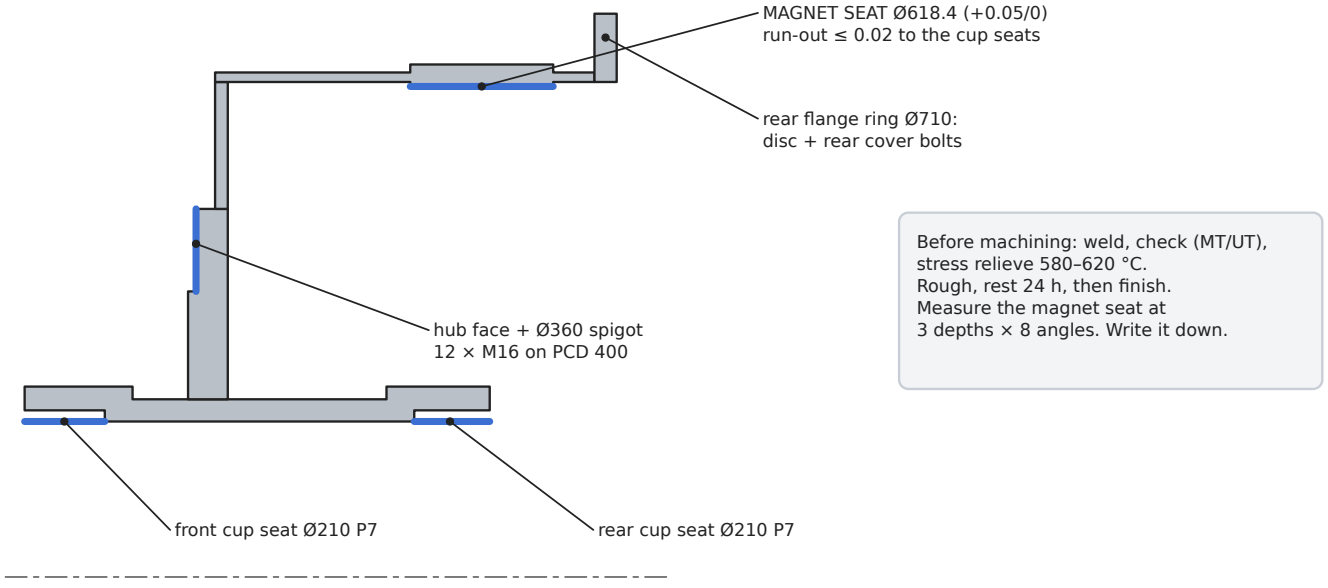
Drawing NP1-210 Rev B (doc 17). Process:

1. Shell: seamless ring-rolled forging, or plate-rolled with a full penetration longitudinal seam (100 % UT to EN ISO 17640, level 2). Raw wall under the magnet zone ≥ 13.8 + 3 mm machining allowance each side.
2. Weld the inner hub web, outer web, bearing tube and rear flange ring in a jig that keeps the axis straight. WPS qualified to EN ISO 15614-1, preheat as required for the thickness, quality level B to EN ISO 5817.
3. NDT: **100 % MT** of all fillet welds, **100 % UT** of full-penetration welds.
4. **Stress relieve**: 580 to 620 °C, 1 h per 25 mm (min 1 h), furnace cool to 300 °C. This is essential for holding 0.02 mm after machining.
5. Shot blast Sa 2.5 outside; prime outside only (the inside is machined).

6.2 R20 Machine the drum in one set-up

G13 Machine the drum in ONE clamping, then measure before unclamping

Section of the welded drum (real shape). Blue = surfaces that must line up within 0.02 mm.



Clamp on the outer web and rough-turn everything, rest 24 h, then finish-machine **all of these in one set-up**:

Feature	Size	Tolerance
Cup seats (front and rear)	Ø210 P7	Coaxial **0.01** to each other
Magnet seat	**Ø618.4**, x 335 to 425	+0.05/0; run-out **≤ 0.02** to the cup seat axis; Ra ≤ 3.2
Shell bore outside the magnet seat	Ø624	
Hub face and spigot	Ø360 f7, face ⊥ 0.02	Hub bolts 12 × M16, PCD 400

Feature	Size	Tolerance
Rear flange ring face	Ø710	Face ⊥ 0.03; disc bolts 24 × M10 on PCD 678, rear cover bolts 24 × M10 on PCD 644

Measure on the machine before unclamping: cup seat diameters and coaxiality, magnet seat diameter at 3 axial × 8 angular positions, run-out. Record the actual **magnet-seat run-out**: it enters the air gap budget (0.020 allowed).

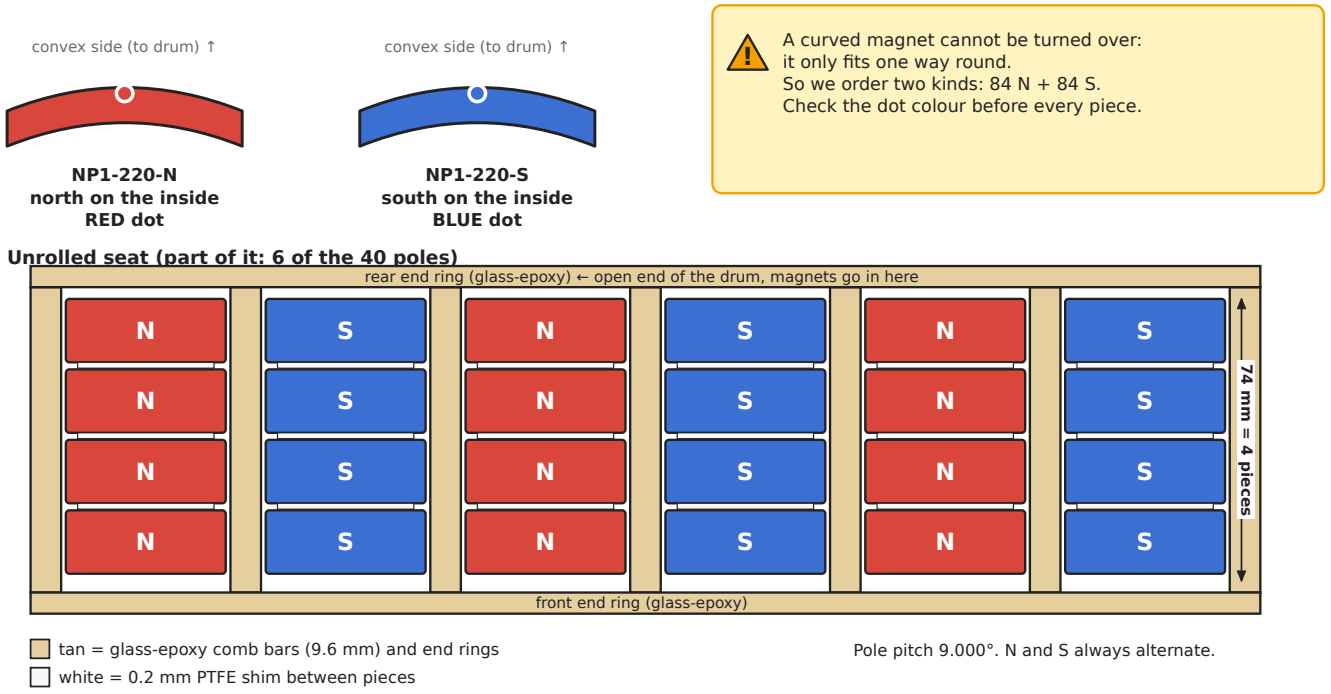
6.3 R30 Pre-balance the bare drum

Balance on a mandrel through the cup seats to **G6.3 at 250 rpm** (residual ≤ 35,700 g·mm for the complete rotor mass of 149 kg; apply proportionally to the bare drum). Correction by drilling the outer web outside the magnet zone. Final balance is done with the hub on the bench (G2.5, ≤ 14,200 g·mm, doc 14). After the magnets are in, the rotor must never go on a balancing machine with steel parts near the drum bore.

6.4 R40 Prepare the magnet seat and fit the pole comb

G14 The magnets: two kinds, and where each one goes

The drum seat is shown unrolled flat, looking from the stator side (as the stator sees it).



1. Mask everything except the magnet seat (x 335 to 425).
2. **Grit blast** the seat with clean alumina, Ra 3 to 6 µm. Blow off, degrease with acetone, then apply the adhesive maker's primer within **4 h** of blasting.
3. Drum on the **non-magnetic rotary table**, axis vertical, **rear end up** (open end; the rear cover is not fitted yet).
4. Bond the **40 comb bars** (GRP, 9.6 × 7.0 × 74 mm) at the pole gaps using an aluminium indexing template located on the hub spigot: pole pitch **9.000°**, bar centre tolerance **±0.05°** (±0.27 mm at r 305). Comb bar 1 at the angle marked on the drum from the hub bolt hole 1.
5. Bond the **front G11 end ring** at x 335 to 343 (closes the front end of the magnet pack).
6. Cure the comb at room temperature for 24 h.

6.5 R50, R60 Receive, sort and bond the magnets (jig NP1-T02)

Receive and sort (in the magnet room):

1. Count against the packing list; check the dot colour of every piece against its variant label.
2. Measure the magnetic moment of every piece in the **Helmholtz coil** (NP1-T09): accept within ±1.5 % of the lot mean, matching the supplier's value within ±1 %.
3. Visual: no chips > 0.5 mm, no coating damage. Chipped pieces are scrapped (section 2.4), not repaired.
4. **Sort into poles:** 4 pieces per pole so that the 40 pole sums are within **±0.5 %** of each other. Then place the poles so that strong and weak poles alternate around the drum. Record the piece serials per pole position.

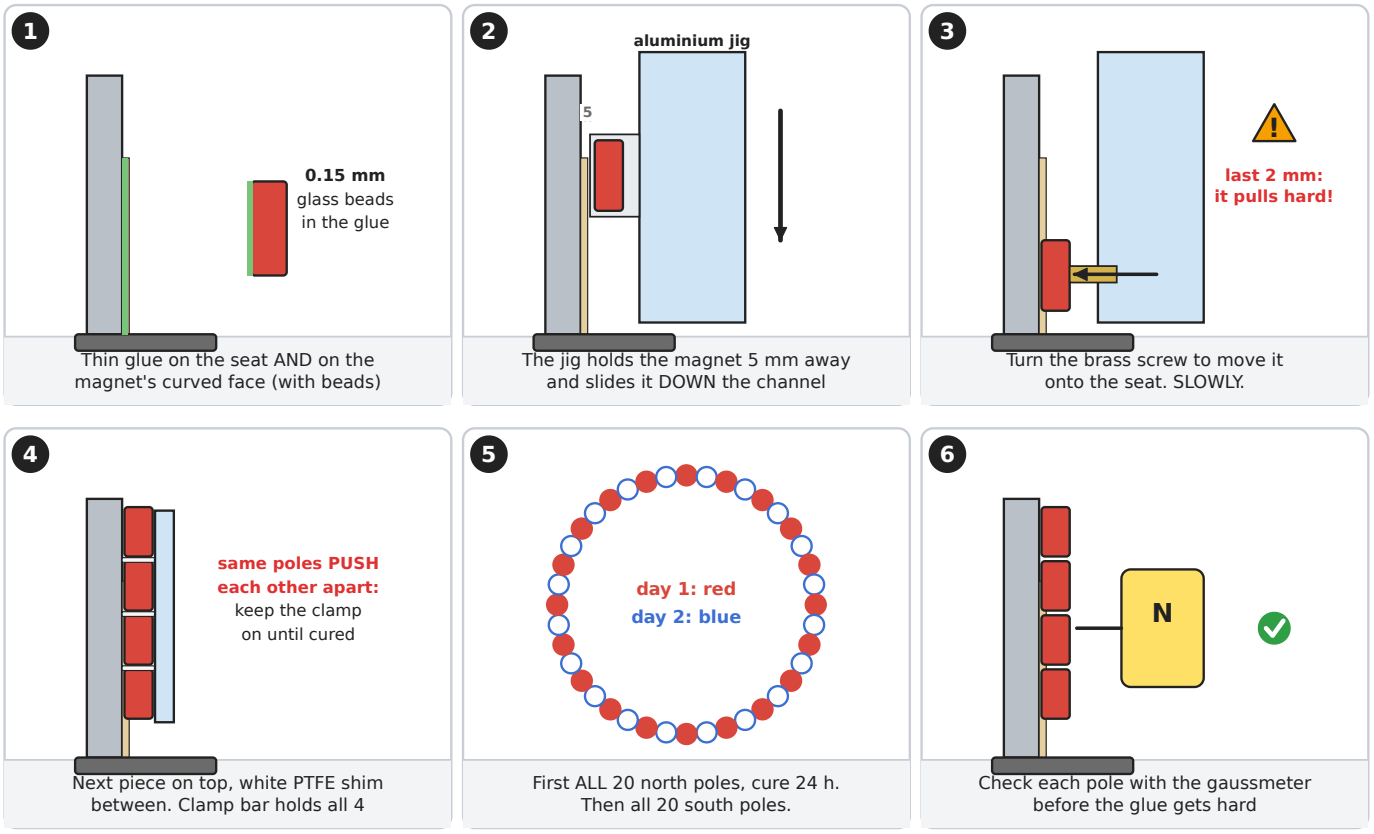
Bond (two people, written procedure signed):

The jig NP1-T02 is an aluminium carriage that runs on a rail along the drum axis, indexed on the comb. It holds one magnet in a POM shoe 5 mm off the seat, slides it axially into its channel and lowers it with a fine-pitch **austenitic stainless or brass screw** onto the adhesive.

1. Mix adhesive with **0.15 mm glass beads** (1 to 2 % by weight); apply a thin, even layer on the seat in the channel, and butter the convex magnet face. Respect the pot life; mix only for 4 poles at a time.
2. Order of poles: **first all 20 N poles** (every second channel), then cure, then **all 20 S poles**. An S piece entering a channel between two N poles is pulled equally from both sides, so the fresh bond carries no side load.
3. In each channel place the **rear-most piece first** (deepest, x 398.8 to 417), against a PTFE stop, then the next three pieces with a **0.2 mm PTFE shim** between pieces. Pieces of the same pole **repel each other axially**: hold each one with the jig's axial clamp bar until the next is down, and leave the clamp bar on the full pole until cure.
4. Lower each piece slowly with the screw; the last 2 mm the pull rises steeply, so the screw must stay in control. Squeeze-out is wiped with a plastic spatula.
5. Check with the gaussmeter that the pole face reads the right polarity (N poles: north on the inside) before the adhesive gels.
6. Cure 24 h at room temperature, remove clamp bars and shims (shims of PTFE come out; any that stick stay, they are non-magnetic).
7. After all 40 poles: fit the **rear G11 end ring** (x 417 to 425), then **post-cure** the drum at **80 °C for 4 h** (ramp ≤ 1 K/min). N40UH is unaffected at 80 °C.

G15 Put the magnets in with the jig (NP1-T02). Two people. Never by hand.

The drum stands upright, open (rear) end up, on the non-magnetic turntable.



After all 40 poles: rear end ring, then warm the whole drum to 80 °C for 4 hours (slow heating). Then scan every pole (section 6.6). Chipped magnet = scrap it in the oil tin, never glue chips.

6.6 R70 Pole map (scanner NP1-T10)

With the drum on the rotary table, scan the radial field 1.0 mm from the magnet faces at mid-stack and at both piece ends, 0.1° steps:

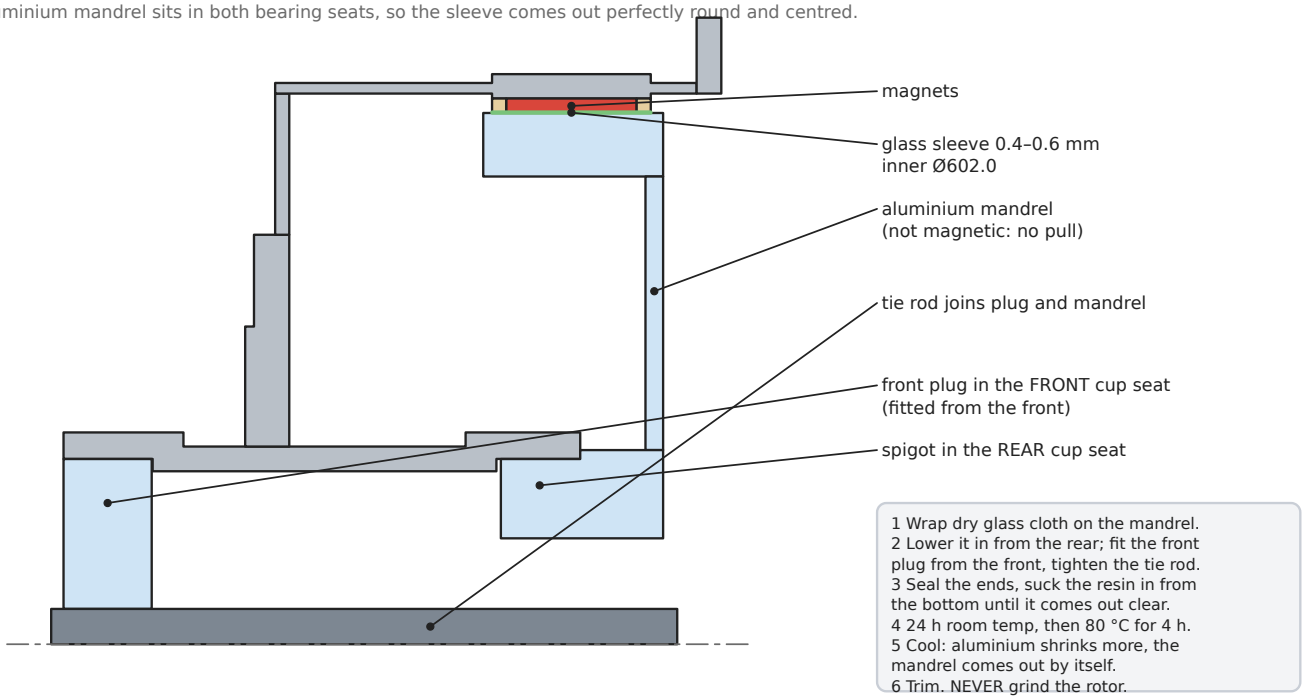
Check	Accept
Polarity sequence	Strictly alternating N/S, 40 poles
Pole peak and flux (integral)	Each pole within **±2 %** of the mean
Pole centre angle	Within **±0.1°** of nominal
Axial uniformity	No dip > 5 % over a piece joint (missing or cracked piece)

Keep the map in the record: it is the reference for the teardown inspection (doc 15, T9).

6.7 R80 Mould the encapsulation sleeve (mandrel NP1-T06)

G16 Mould the glass sleeve over the magnets (mandrel NP1-T06)

The aluminium mandrel sits in both bearing seats, so the sleeve comes out perfectly round and centred.



The sleeve closes the magnet pack and defines the rotor bore.

1. The mandrel is **aluminium** (non-magnetic, no pull), Ø**602.0 +0/−0.02** at 20 °C, release-coated, with two precision spigots that locate in the **front and rear cup seats** of the drum. Its surface is therefore concentric with the bearing axis within the mandrel's own run-out (≤ 0.01).
2. Wrap the dry glass fabric on the mandrel (layup set on Q1 so the cured sleeve is 0.4 to 0.6 mm everywhere), plus a 5 mm overlap onto each G11 end ring.
3. Lower the mandrel into the drum (axis vertical) with the crane, spigots into the cup seats.
4. Seal the ends; **vacuum infuse** the resin from the bottom, vent at the top until resin flows clear.
5. Cure 24 h at room temperature, then post-cure **80 °C, 4 h**. Cool the assembly: aluminium shrinks more than steel, so the mandrel releases by itself; pull it straight out with the crane.
6. Trim flash at the ends. **Nothing on the rotor may be ground or machined after the magnets are in.**

6.8 R90 Complete the rotor

1. **Inspect the sleeve bore**: diameter at 3 axial × 8 angular positions, **Ø602.0 +0.05/−0.02**, run-out to the cup seats **≤ 0.03**; tap-test for voids (none > 10 mm). Record.
2. **Press both cups** into the drum (cups cooled to −30 °C or drum warmed to 60 °C, do not exceed 80 °C), check seating with a 0.03 feeler.
3. Fit the **rear seal cover NP1-211** with its 195 × 220 × 12 lip seal onto the drum housing (Rev B order, doc 17).
4. Slide the **brake disc NP1-230 on from the front** over the Ø646 shell (Rev B), 24 × M10 10.9 on PCD 678, torque per drawing.
5. Fit the magnet-zone cover: a POM ring that closes the drum bore for storage, with the label **"Magnetised rotor, strong field inside"**.
6. Weigh (≈ 149 kg with disc and covers; record).

7. Interfaces to check before joining

Interface	Check	Value
Stator tip OD to rotor sleeve bore	Calculated gap from measured diameters	**2.0 mm** nominal; minimum calculated radial gap at any point ≥ **1.80 mm** (budget 0.20 worst case, doc 14)
Stator bolting face to carrier rim axis	Record from S50	Within 0.02
Cable path	Leads and PT100 through the Ø28 holes	No sharp edges; grommets fitted
Rear seal cover and disc on the drum	Rev B	As section 6.8

If the calculated minimum gap is below 1.80 mm, stop and review with Category 4 before joining.

8. Joining stator and rotor (assembly bay, magnet hazard)

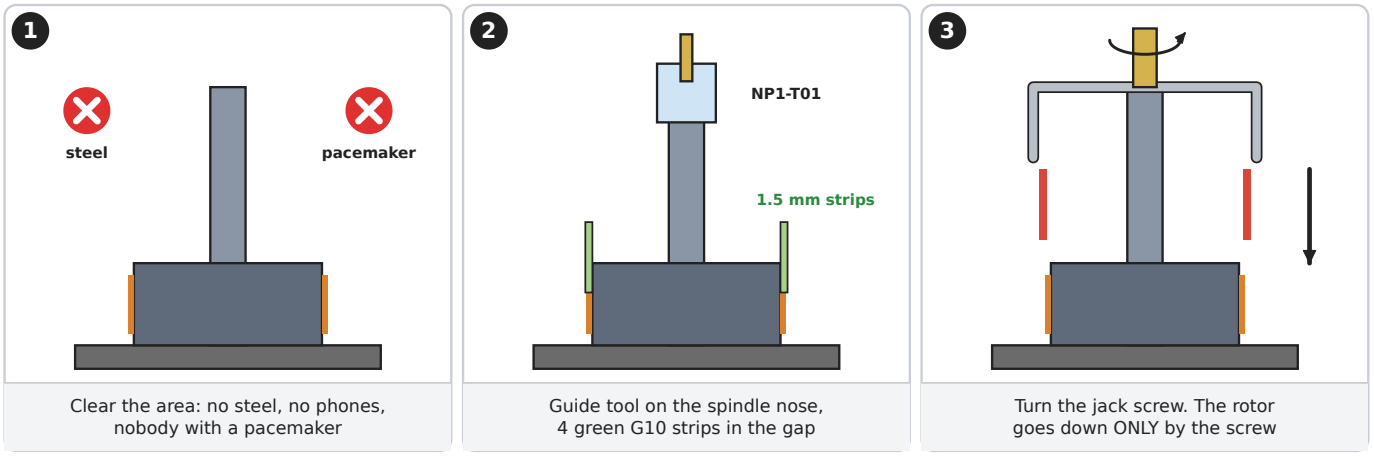
This follows the Rev B order in doc 17 (§2.1). Steps 1 to 3 have no magnet hazard; step 4 is the critical step.

Step	Work
J10	**Spindle prepared** (doc 14 §7.2 step 3): cooling insert shrunk in, PT100s under the cones. Spindle **vertical, nose up**, flange bolted to the assembly stand
J20	**Stator onto the bare spindle**: lower the stator over the nose onto the flange, leads through the Ø28 holes. Thermal paste on the bolting face, 2 dowels, **12 × M12 10.9, 110 Nm** (doc 14)
J30	**Seal sleeve NP1-102** with O-ring onto the Ø155 shoulder; **rear cone** pressed against the sleeve and shoulder (cone heated to 100 °C max, induction heater)
J40	**Rotor over the stator** with guide NP1-T01 (below)
J50	**Front bearing**: front cone, spacer NP1-101, KMT 27 nut; set end play **−0.02 to −0.05 mm**; record rolling torque; **then** the front seal cover (doc 17)
J60	**Rear cover** NP1-212 with the Ø400 seal over the flange; breather; measure the air gap

8.1 J40 Lowering the rotor (written procedure, two people plus a supervisor)

G17 Put the rotor over the stator: the most dangerous step

≈ 27 kN of magnet pull. If the rotor goes 1 mm off centre, it pulls 1.9 kN more to that side.



Clear the area: no steel, no phones, nobody with a pacemaker

Guide tool on the spindle nose, 4 green G10 strips in the gap

Turn the jack screw. The rotor goes down ONLY by the screw

A strip gets tight? STOP. Check the guide. Never push the rotor sideways

Rear bearing sits on its cone: take the load off, pull out the strips

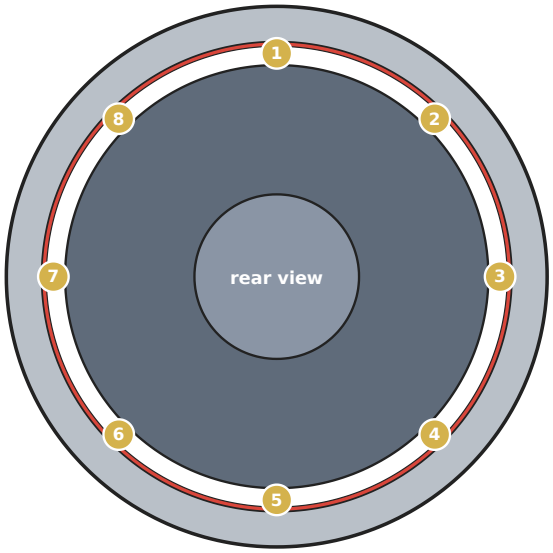
Turn it one full turn by hand: free, smooth, no rubbing

1. Clear the zone: no loose steel within 2 m, pacemaker check, all tools non-magnetic (NP1-T11).
2. Fit the guide tool NP1-T01 on the spindle nose. Its guide sleeve centres the drum through the front cup seat; the screw jack carries the rotor.

3. Insert **four G10 feelers 1.5 mm** at 90° into the gap entry (tied with cord so they can be pulled out later).
4. Lift the rotor (rear end down, magnets towards the stator) with textile slings, hand it over to the screw jack above the guide.
5. Lower **by the screw only**. Watch the gap at the four feelers. If any feeler becomes tight, **stop** and check the guide; never push sideways by hand. The rotor is self-centring only while the guide holds it; any offset is pulled further at 1.9 kN per mm.
6. When the rear cup is seated on the rear cone, remove the jack load, pull out the feelers.
7. Turn the rotor slowly by hand through one full turn: it must turn freely with only the bearing drag and a smooth cogging feel (cogging ≈ 0.85 Nm peak-to-peak, barely noticeable). Any rub: lift and inspect.

8.2 J60 Air gap measurement

G18 Check the air gap at 8 places × 2 ends (brass or plastic feelers only)



1 Look through the inspection port.
2 Slide a brass feeler between the sleeve and a tooth tip. Write the size.
3 Turn the rotor 45°, repeat to 8.
4 Do it at the front end and rear end.

Every reading: ≥ 1.80 mm
Biggest − smallest: ≤ 0.20 mm

pos	1	2	3	4	5	6	7	8
front	___	___	___	___	___	___	___	___
rear	___	___	___	___	___	___	___	___

Through the inspection port in the rear cover, with **brass or plastic** feeler gauges, at **8 positions × 2 axial ends** (turning the rotor by 45° steps), measure the gap between the sleeve and the tooth tips:

Accept	Value
Every reading	≥ **1.80 mm**
Max − min (eccentricity × 2)	≤ **0.20 mm**

Record. Then fit the plug in the port.

9. Generator tests after joining

Done on the assembly stand (no rotation needed) and then on the bench (doc 15). These are the generator-specific acceptance values.

Test	When	Accept
Phase resistance, IR, PI	After joining	As 5.13
AC hipot	Not repeated; if required, 80 % (1.8 kV)	
Hand-turn back-EMF polarity	Rotor turned slowly by hand, scope on the phases	Positive sequence defined for the design rotation, three equal phases
Back-EMF (doc 15, T2)	Bench, 50, 100, 178 rpm, magnets at known temperature	Line EMF **≈ 2.42 V/rpm at 20 °C** (≈ 430 V at 178 rpm; scaled value, replace with the FEA rerun) within **±3 %**; phases within ±1 %; THD ≤ 1.5 %
Cogging (T2)	Bench, slow turn, torque transducer	≤ 0.5 % of rated (≤ 4 Nm)
Short circuit and demag check (T3)	Bench	EMF change after test ≤ 0.5 %
Load and efficiency (T4)	Bench	η ≥ 93 % at rated
Thermal (T5)	Bench	Hot spot ≤ 130 °C at 110 %

10. Q1 process qualification (before P1 parts are committed)

Doc 04 already asked for a stator segment trial. With the segmented design this becomes a short, decisive qualification. Build in weeks 3 to 8 while the magnets are on order.

Article	Content
Q1-S stator sector	**5 wound segments** (one full group +A −A +A plus one neighbour each side) in a 40° carrier sector of real material, liners, wedges, connection joints, 2 PT100s, VPI and potted like P1
Q1-M magnet panels	3 flat steel panels with grit blast + primer, each with 4 unmagnetised test pieces bonded and sleeved as P1; 12 single lap-shear coupons
Q1-T motorettes	6 twisted-pair and coil motorettes in the P1 insulation system (IEC 60034-18-21/-41)

Test	Accept
Winding: all 5 coils wound to the layer plan, coil fit into the sector without force	Yes; record wind time per coil
Coil resistance vs calculated	Sets the P1 acceptance mean
VPI: section-cut one segment after cure, inspect slot fill	Resin-filled, voids < 5 % of slot area, no dry areas between strands; sets uptake and capacitance-rise targets
Potting: 50 thermal cycles −40 / +150 °C	No cracks, no debonding, IR ≥ 1,000 MΩ after cycling
PDIV and RPDIV on motorettes, dry and at 90 % RH	Confirms the 1.9 kV peak target
Thermal endurance on motorettes (short form)	No early failure versus class H rating
Magnet bond lap shear at −40, 23, 120 °C, after 50 cycles	≥ 15 / 15 / 5 MPa
Salt-fog on Q1-M panel, 1,000 h (ISO 9227 NSS) with a scribed sleeve	No corrosion under the sleeve beyond 2 mm from the scribe
Sleeve thickness by section	0.4 to 0.6 mm

Q1 results go into this document as revision B (targets replaced by measured values) before the P1 winding starts.

11. Records, traceability and repair

Generator record (one per serial):

- Material certificates: steel coil, wire lot, liner, resin and potting lot, magnet lot with per-piece moment, adhesive lot, drum and carrier steel.
- Segment traveller: serial, sheet count, height, tip radius, coil test results, position on the carrier.
- Stator: run-out map, pre-VPI and final tests, VPI log (vacuum, pressure, temperatures, uptake), potting log.
- Rotor: drum machining record, balance, pole sort table, bonding log (who, when, adhesive lot, room temperature and humidity), pole map, sleeve bore map.
- Joining: gap readings, bearing end play, rolling torque, torques.

Repair rules:

Fault	Before VPI	After VPI
Coil failed test	Pull the segment, replace with a spare	Strip the stator (burn-off oven not allowed with bonded segments; chemical/solvent strip) and re-wind affected segments, or replace the stator. This is why all coil tests happen before VPI
Magnet cracked or misplaced	Before sleeve: remove by heating the bond to 150 °C locally and replace	After sleeve: return the rotor to the magnet room, cut out the sleeve over the pole, replace, re-sleeve that section and repeat the pole map
Sleeve void	Fill by injection with sleeve resin	Same

12. Open items

#	Item	Owner	Due
1	~~Decision on the stator build route (M1)~~ **Decided 2026-11-03: segmented stator**	Martin	Done
2	FEA rerun with segment joints and tolerance scatter; freeze turns (27 or 28) and the expected EMF	Cat. 4	Week 1
3	Magnet eddy-loss check, then place the magnet order	Cat. 4	Week 1
4	Drawings NP1-200 Rev C, NP1-201 Rev C (segment DXF), tools T03 to T10, connection ring	Cat. 4/10	Weeks 1 to 3
5	Choose resin, potting compound and adhesive products; get maker approval for the wire enamel and magnet coating	Cat. 10	Week 2
6	Q1 qualification, revise this document to revision B	Cat. 10/11	Weeks 3 to 8
7	Update doc 12 for the worst-case voltage with the frozen turn count	Cat. 7	After item 2
8	Series route (stamping die, larger VPI batches, automated winding) after P1	Cat. 10	After P1

Appendix A. Coil connection plan

Teeth are numbered 1 to 45 in the direction of rotation, tooth 1 at the reference mark on the carrier (12 o'clock, aligned with dowel 1). Full table with series position: `coil_connection.csv`.

Teeth	Phase and sense
1, 2, 3	+A, −A, +A
4, 5, 6	+C, −C, +C
7, 8, 9	+B, −B, +B
10 to 18, 19 to 27, 28 to 36, 37 to 45	Same 9-tooth pattern repeated (5 repeats)

Phase A in series: 1 → 2 (reversed) → 3 → 10 → 11 (reversed) → 12 → 19 → 20 (reversed) → 21 → 28 → 29 (reversed) → 30 → 37 → 38 (reversed) → 39, then the star point. Phase B starts at tooth 7, phase C at tooth 4, each with the same pattern. Lead U = phase A start (tooth 1), V = phase B start (tooth 7), W = phase C start (tooth 4). Confirm the phase sequence at the hand-turn EMF test and swap V/W at the terminal if the design rotation gives negative sequence.

Appendix B. Key numbers for the shop

From manufacturing_summary.json:

Item	Value
Slot width top / bottom / opening	18.9 / 9.8 / 4.0 mm
Free area per coil side / copper	425 / 211 mm² (fill 0.50)
Turn	4 × Ø1.55 mm = 7.55 mm², 3.07 A/mm² at 23.2 A
Mean turn length	227 mm
Wire per coil	6.6 m per strand; ≈ 1,200 m of Ø1.55 for the stator
Copper	20.1 kg in the winding; order 25 kg
Resistance at 20 °C	Coil 14.5 mΩ, phase 0.217 Ω, line to line 0.435 Ω
Copper loss at 75 °C, rated	≈ 430 W
Segment sheets	205 per segment, 9,225 in total (+5 %)
Magnet piece	37.86 × 18.35 × 7.70 mm arc, 40.9 g, 6.55 kg total
Pull of one piece on steel	≈ 280 N
Pole gap (comb bar width) at r 305	9.6 mm
Expected line EMF, 178 rpm	≈ 392 V at 100 °C, ≈ 430 V at 20 °C (28 turns, scaled)
Worst-case peak, 250 rpm, −40 °C	≈ 920 V (28 turns) / ≈ 890 V (27 turns)
Hipot / repeat / surge / PDIV	2.3 kV / 1.8 kV / 3.5 kV peak / ≥ 1.9 kV peak
Balance at 250 rpm	G6.3 ≤ 35,700 g·mm (bare drum stage), G2.5 ≤ 14,200 g·mm (final)

Appendix C. Stator build routes compared (decision M1, route A chosen)

	A. Segmented stator (recommended)	B. Open slots + magnetic wedges	C. As drawn, drop-in by hand
Core	45 bonded tooth segments in dovetail grooves	One-piece bonded ring, tooth tips removed (slot open 18.9 mm)	One-piece bonded ring as doc 14
Coil	Precision layer-wound on each tooth, fill 0.50 achievable	Pre-wound on a former, slid over the parallel tooth; fill 0.50 achievable	Fed wire by wire through 4 mm; ≈ 0.40 fill at best
Electromagnetic effect	Same shape; joints cost < 1 % EMF (FEA rerun)	Without tips the gap flux and cogging change; magnetic wedges (μr ≈ 3 to 5) recover most of it; FEA rerun needed	Copper −20 %: current density ≈ 3.8 A/mm², copper loss ≈ +25 %, hot spot ≈ +10 K, η about −0.5 %; or re-design slots
Air gap control	Segment tolerance + groove machining (±0.05 total)	One-piece ring, best run-out	One-piece ring
Testability, repair	Every coil tested before assembly; segment swap before VPI	Coils tested before fitting	Faults found only on the full stator
Long-term risk	Joint fretting prevented by VPI and clamp ring	Magnetic wedges can loosen and break under vibration over 20 years	Insulation damage while forcing wires in
Drawing changes	Lamination, carrier	Lamination, wedges	None

Route A is the industry standard for concentrated-winding machines that need high fill, and it fits the quality-first rule: every coil is the same, tested, and the tightest dimensions are machined into a steel carrier, not built up by hand.